

Figure 1: *targetcli* interactive mode shown with *ls* command

connectable storage or output device uniquely identified on a SCSI bus. In iSCSI, in which the SCSI bus is emulated across an IP network, a target can be a dedicated physical device in a network attached storage (NAS) enclosure or an iSCSI software configured logical device on a networked storage server. Targets like HBAs and initiators are end points in SCSI bus communication, passing command descriptor blocks to request or provide storage transactions.

To provide access to the storage or output device, a target is configured with one or more logical unit numbers (LUNs). In iSCSI, LUNs appear as the target's sequentially numbered disk drives, although targets typically have only one LUN. An initiator performs SCSI negotiations with a target to establish a connection to the LUN, which responds as an emulated SCSI disk block device, which can be used in raw form or formatted with a client supported file system.

Warning: Do not mount single system file systems to more than one system at a time. iSCSI allows shared target and LUN access from multiple initiator nodes, requiring the use of a cluster capable file system such as GFS2. Mounting standard file systems designed for local single system access (e.g., ext3, ext4, FAT32, XFS and ZFS) from more than one system concurrently will cause file system corruption.

iSCSI provides for LUN masking by using ACLs to restrict LUN accessibility to specific initiators, except when shared access is intended. ACLs can ensure that only a designated client node can log in to a specific target. On the target server, ACLs can be set at the TPG level to secure groups of LUNs, or can be set individually per LUN.

iSCSI target configuration

Let's go through a demonstration of target server configuration.

targetcli is both a command line utility and an interactive shell in which to create, delete and configure iSCSI target components. Target stacks are grouped into a hierarchical tree of objects, allowing easy navigation and contextual configuration. Familiar Linux commands are used in this shell. These include *cd*, *ls*, *pwd* and *set*.

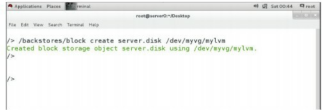


Figure 2: A block has been created for the logical volume */dev/myvg/mylv*



Figure 3: IQN has been created for the target

The *targetcli* shell also supports TAB completion. An administrator can use TAB to either complete partially-typed commands or view a list of acceptable keywords at the current location in a command.

Configuring a target server

Install *targetcli* by using the following command:

```
# yum -y install targetcli.
```

Run *targetcli* with no options to enter interactive mode:

```
# targetcli
```

Create backing storage (backstores). There are several types of backing storage, as listed below:

- **Block:** A block device is defined on the server. It is a disk drive, disk partition, a logical volume, multi-path device, or any device files defined on the server that are of Type b.
- **fileio:** This creates a file of a specified size in the file system of the server. This method is similar to using image files to be storage for virtual machine disk images.
- **pscsi:** Physical iSCSI, this permits pass through to a physical SCSI device connected to the server. This backstore type is not typically used.
- **Ramdisk:** This creates a ramdisk device of a specified size, in memory on the server. This type of storage will not store data persistently. When the server is rebooted, the ramdisk definition will return when the target is instantiated but all data would have been lost.

Note: For such cases, a logical volume has been created: */dev/myvg/mylv*, although a simple partition can also be created.

Examples include using an existing logical volume, a disk partition, and a new file at a specified size. The backstores get displayed as deactivated.